higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL WORKMANSHIP
NQF LEVEL 3

(12041013)

7 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

Copyright reserved

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCNO

QUESTION 1

1.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Planning and organising tasks help to prevent mistakes.

1.1.2 Hazards are clearly identified when using work permits.

1.1.3 Verification of isolation is not necessary when disconnecting electrical supplies.

1.1.4 A task list is a tabulated notice informing employees of what must be done every day, week or month.

1.1.5 Bar charts give workers time frames in which to complete tasks.

(5 × 1) (5)

1.2 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.2.1–1.2.4) in the ANSWER BOOK.

1.2.1 The term used for an unsafe condition in the workplace is …

A accident.
B unsafe act.
C work factor.
D personal factor.

1.2.2 The term used for an unsafe condition or act that could have caused injury to people in the workplace is ...

A injury.
B incident.
C accident
D fatality.

1.2.3 The term used which describes the type of working environment in which people work is ...

A good lighting.
B bad ventilation.
C slippery floors.
D physical condition.

1.2.4 The term used for a cluttered, dirty, unorganised workshop is ...

A bad housekeeping.
B bad maintenance.
C bad planning.
D bad attitudes.

(4 × 1) (4)

Copyright reserved

1.3 State THREE factors which may cause stress to people in the workplace. (3)

1.4 State TWO functions of health and safety inspectors. (2)

1.5 Employers and employees must always put the health and safety of all people in the workplace first.

State THREE ways in which the following people can improve health and safety in the workplace:

1.5.1 Employers

1.5.2 Employees

(2 × 3) (6)

1.6 State TWO of the procedures used when recording and investigating incidents. (2)

1.7 State THREE conditions that cause arm, leg, hand, neck, back and foot injuries when working in confined spaces. (3)

[25]

QUESTION 2

2.1 Testing and measuring instruments are classified in different ways, because of the effects of voltage and current in their operations.

Name TWO of these classifications. (2)

2.2 Name the test instruments used to measure or test each of the following electrical quantities or devices:

2.2.1 The effectiveness of an earth-leakage relay unit

2.2.2 The amount of current flowing through an electrical circuit

2.2.3 The potential difference between two electrical supply point terminals

(3 × 1) (3)

2.3 Explain TWO factors to consider when caring for testing and measuring instruments. (2)

2.4 Draw ONE fully labelled circuit diagram in which the following test instruments are directly connected:

- A voltmeter
- An ammeter
- A frequency meter

(5)

Copyright reserved

2.5 A voltage transformer has been installed at a site. The full-scale deflection of the voltmeter is 20 V and the VT transformer has 1 900 turns on the primary winding and 100 turns on the secondary winding respectively.

Calculate the expected line or primary voltage. (3)
[15]

QUESTION 3

3.1 Explain the operation of a convection heater. (3)

3.2 Draw the basic circuit diagrams of a three-heat stove switch. Only show how the elements are connected in the low-, medium- and high-switch positions (excluding the switch). (3)

3.3 Explain the operation of the switch used in an automatic kettle. (3)

3.4 Briefly explain the basic function of each of the following components:

3.4.1 A washing machine door or lid switch

3.4.2 An oven thermostat

3.4.3 A bimetal contact or strip

(3 × 1) (3)

3.5 Explain the regulation concerning protective isolation devices used for stoves and geysers. (3)
[15]

QUESTION 4

4.1 Give TWO reasons for testing electrical appliances and power tools. (2)

4.2 Appliances and power tools are classified according to the type of insulation used by the manufacturer.

Name FOUR types of insulation used in Class I and Class II appliances and power tools. (4)

NCV

4.3 Choose a possible fault and repair from COLUMN B that match the fault condition in COLUMN A. Write only the letter (A–G) next to the question number (4.3.1–4.3.4) in the ANSWER BOOK.

COLUMN A		COLUMN B	
4.3.1	A kettle does not boil, but the power indication light is on	A	blown fuse, replace fuse
4.3.2	No steam comes out of the iron	B	the iron power cord is faulty, replace the power cord
4.3.3	One stove plate does not work	C	the kettle switch is faulty, replace switch
4.3.4	Geyser water not hot enough	D	steam vents blocked, clean or replace air vents
		E	timer faulty, replace timer
		F	the kettle element is faulty, replace the element
		G	thermostat setting incorrect, adjust the thermostat

(4 × 2) (8)

4.4 Name ONE fault-finding procedure used when repairing appliances and power tools. (1)

[15]

QUESTION 5

5.1 Name ONE type of bearing used in electric motors. (1)

5.2 Draw the fully labelled circuit diagram showing ONE method of reversing the rotational direction of a capacitor-start induction-run motor. (6)

5.3 State TWO methods used to control the speed of universal motors. (2)

5.4 Motors are classified according to the type of motor enclosures used.

Explain the term *pipe-ventilated* protection. (3)

5.5 5.5.1 What does D.O.L stand for? (1)

5.5.2 Give a reason for using this electrical device or circuit. (2)

[15]

QUESTION 6

6.1 State TWO advantages of DC SMAW machines over AC machines. (2)

6.2 Explain the following terms:

6.2.1 Chipping

6.2.2 A contour

6.2.3 A grooved angle

(3 × 1) (3)

6.3 The chemical flux coating of electrodes has a few functions.

Explain TWO reasons why welding electrodes or rods are coated with flux. (2)

6.4 Explain how you would strike an arc and weld a joint when using the scratch start technique. (4)

6.5 Which TWO tools would you use to clean a welded joint? (2)

6.6 State TWO factors to consider when preparing a workpiece for welding. (2)

[15]

TOTAL: 100

NCED

Copyright reserved